The codes in this repository are based on the Fernández (2012) wavelet approach and were originally developed in 2020, but uploaded to GitHub at a later date. The wavelet decomposition script uses MODWT with the LA8 filter. This project has been forked from the MODWT-MARS repository by 0zean and extended to include multiple wavelet correlation (MWC) analysis for 11 companies listed on the Tehran Stock Exchange.

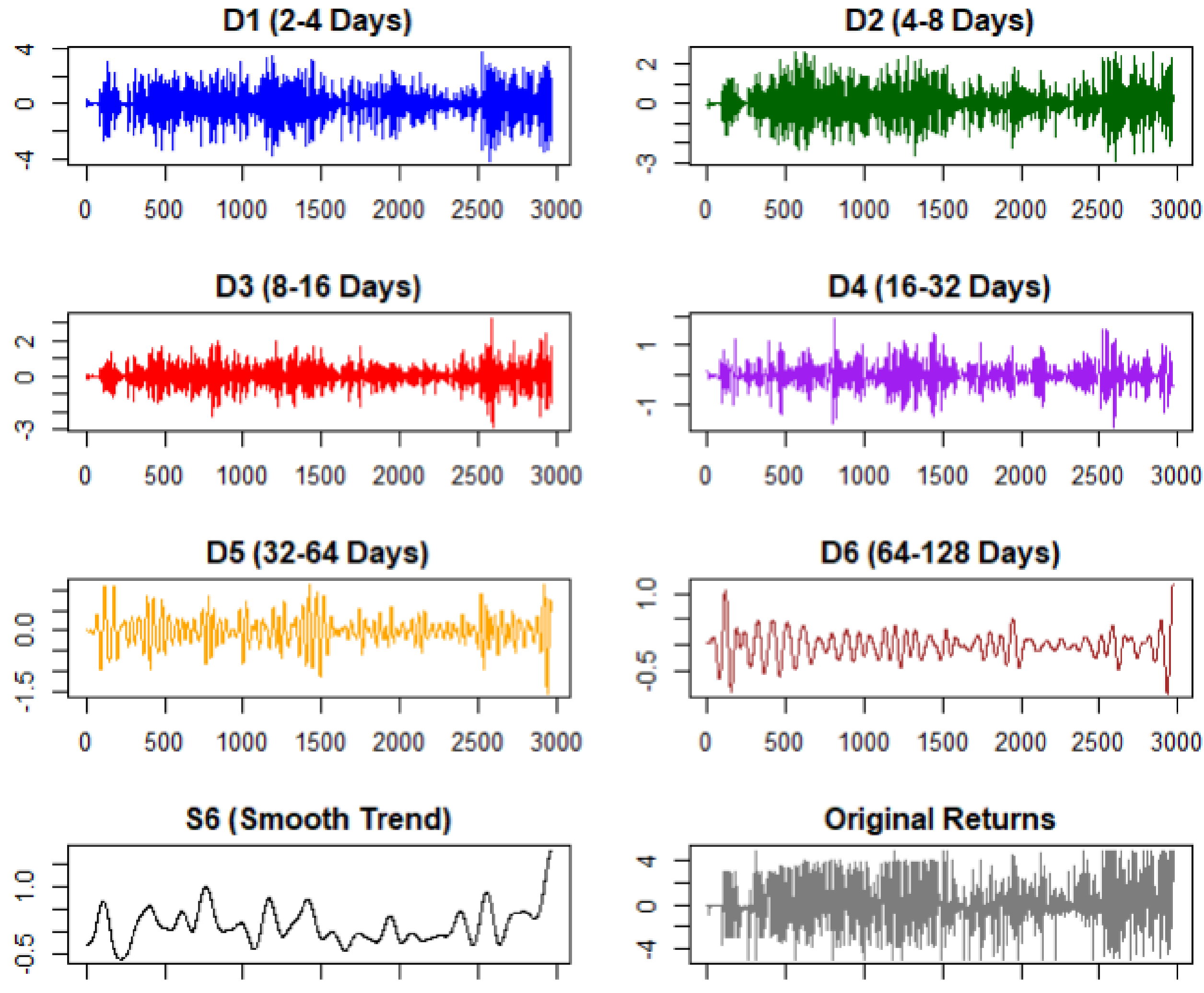


Figure 1: Wavelet Decomposition Output (D1–D6 & S6)
MODWT decomposition of the company returns into 6 detail levels (D1–D6) representing 2–128 day cycles, plus the smooth trend component (S6). (This code is forked from MODWT-MARS)

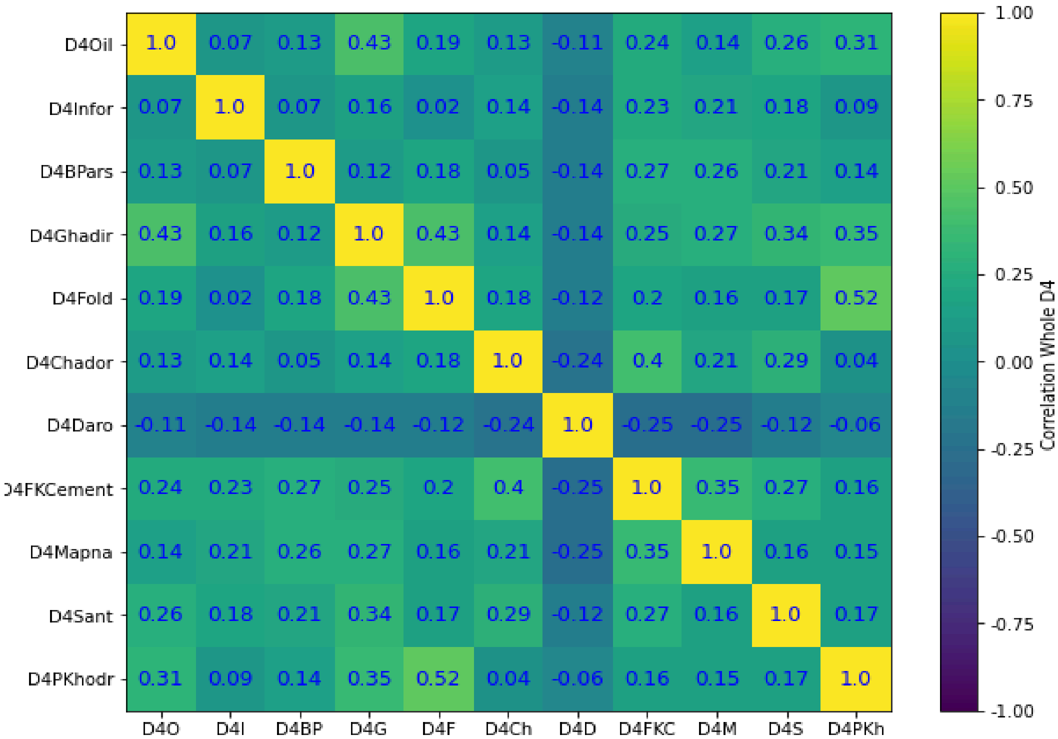


Figure 2: 11×11 Pairwise Correlation Matrix at D4 Level
Pairwise correlation coefficients between 11 companies at the D4 time scale (16–32 days), showing relationships ranging from -1.0 to 1.0.

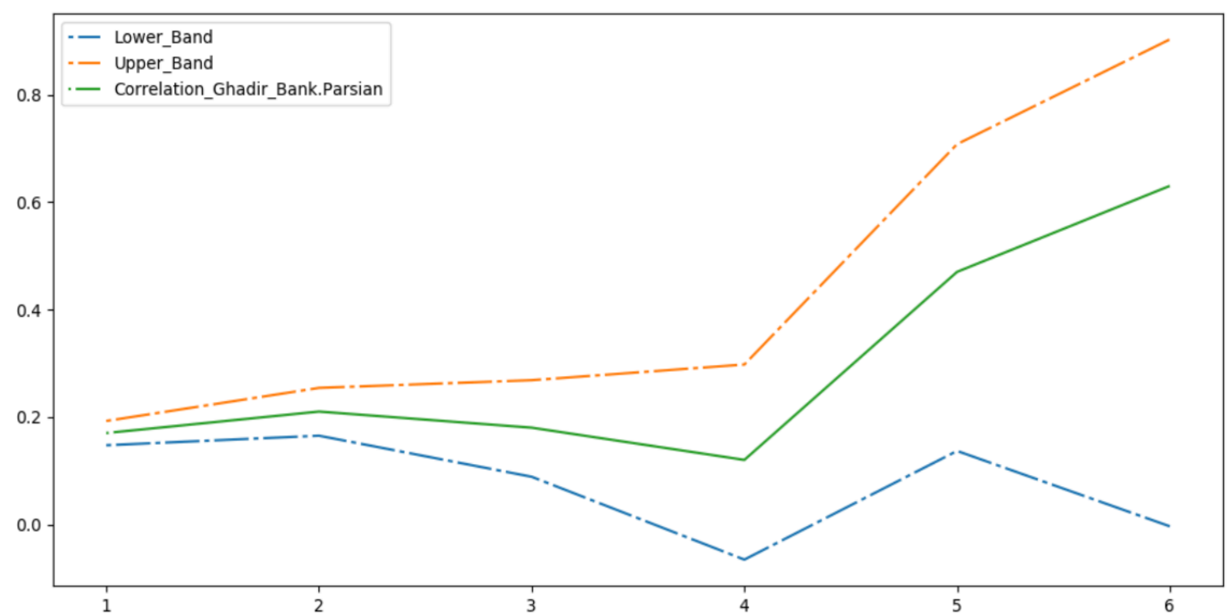


Figure 3: Correlation,Co-Movement, of Ghadir with Bank Parsian (D1–D6)
Correlation between Ghadir and Bank Parsian across 6 time scales, with upper and lower confidence bands.

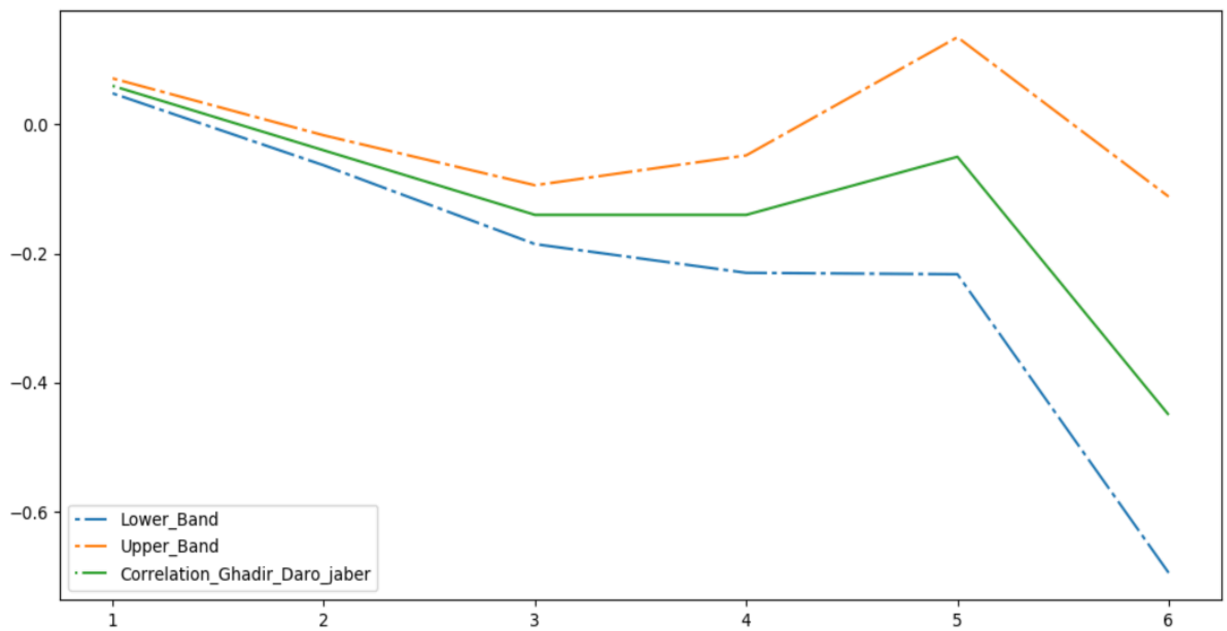
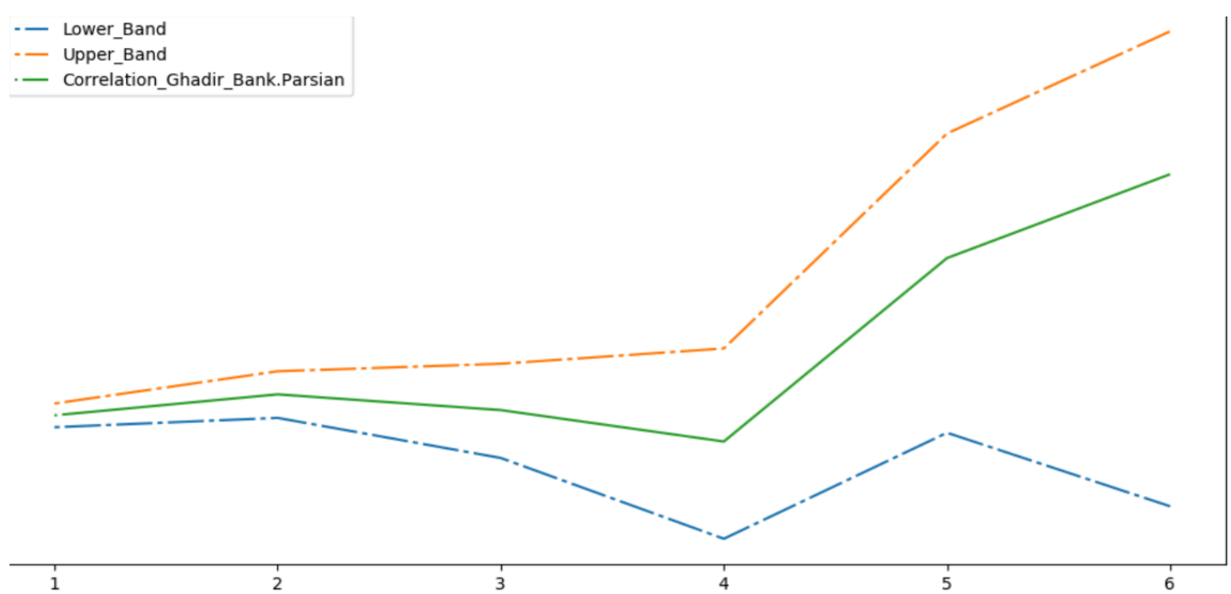


Figure 4: Correlation,Co-Movement, of Ghadir with Daro Jaber (D1–D6)
Correlation between Ghadir and Daro Jaber across 6 time scales, showing negative correlations at higher levels.



میزان هم حرکتی (همبستگی) سطح یک تا شش هولدینگ غدیر با بانک پارسیان شکل (۴-۱۶)

Figure 5: MWC Plot with Confidence Bands
Multiple Wavelet Correlation (MWC), Co-Movement, values for D1–D6 with 99.7% confidence bands, showing increasing correlation at longer time scales.

This project includes five scripts which are designed based on the framework proposed by Fernández (2012). According to this approach, financial time series should be studied at different time scales because market behaviour is not the same in the short term and the long term. Therefore, to conduct a more accurate analysis, the return series of 11 companies should be decomposed into different frequency components, and correlation should be measured separately at each time scale.

Script 1 (R):
This script uses the Maximal Overlap Discrete Wavelet Transform (MODWT) with the LA8 filter to decompose the return series of each of the 11 companies into 6 frequency levels (D1 to D6) and one smooth component (S6). This is done according to the Fernández approach so that correlation can be examined at different time horizons.

Script 2 (Python):
This script calculates the 11×11 pairwise correlation matrix for the D4 level. Pairwise correlation measures the relationship between each pair of the 11 companies at a specific time scale. It also generates a heatmap of these correlations and saves the correlation of one specific company (Ghadir) with the other 10 companies.

Script 3 (Python):
This script calculates a 22×22 correlation matrix which includes 11 series of D6 wavelet coefficients (from the wavelet decomposition of the 11 companies) and 11 series of original returns (from the same 11 companies). It then extracts the 11×11 submatrix (original returns versus D6), calculates its inverse, and obtains the diagonal elements. The maximum diagonal value is used for calculating multiple correlation in the next step.

Script 4 (Python):
This script calculates the Multiple Wavelet Correlation (MWC) for all 6 levels using the formula $MWC = \sqrt{(1 - 1/max\_diag)}$. Unlike pairwise correlation, which measures the relationship between two companies, MWC measures the relationship between one company's returns and a combination of all 11 companies. This indicator shows how much of each company's variation can be explained by the whole market.

Script 5 (Python):
This script uses the Fisher transformation to calculate 99.7% confidence bands for each level and plots the MWC values together with upper and lower bounds. This helps to identify which time scales have statistically significant multiple correlation.

In this way, these five scripts gradually implement the Fernández approach for 11 stock market companies: from wavelet decomposition and pairwise correlation calculation to multiple correlation measurement and final plotting with confidence bands.